2D CAM FOR CNC · IN THE BROWSER

# COPO

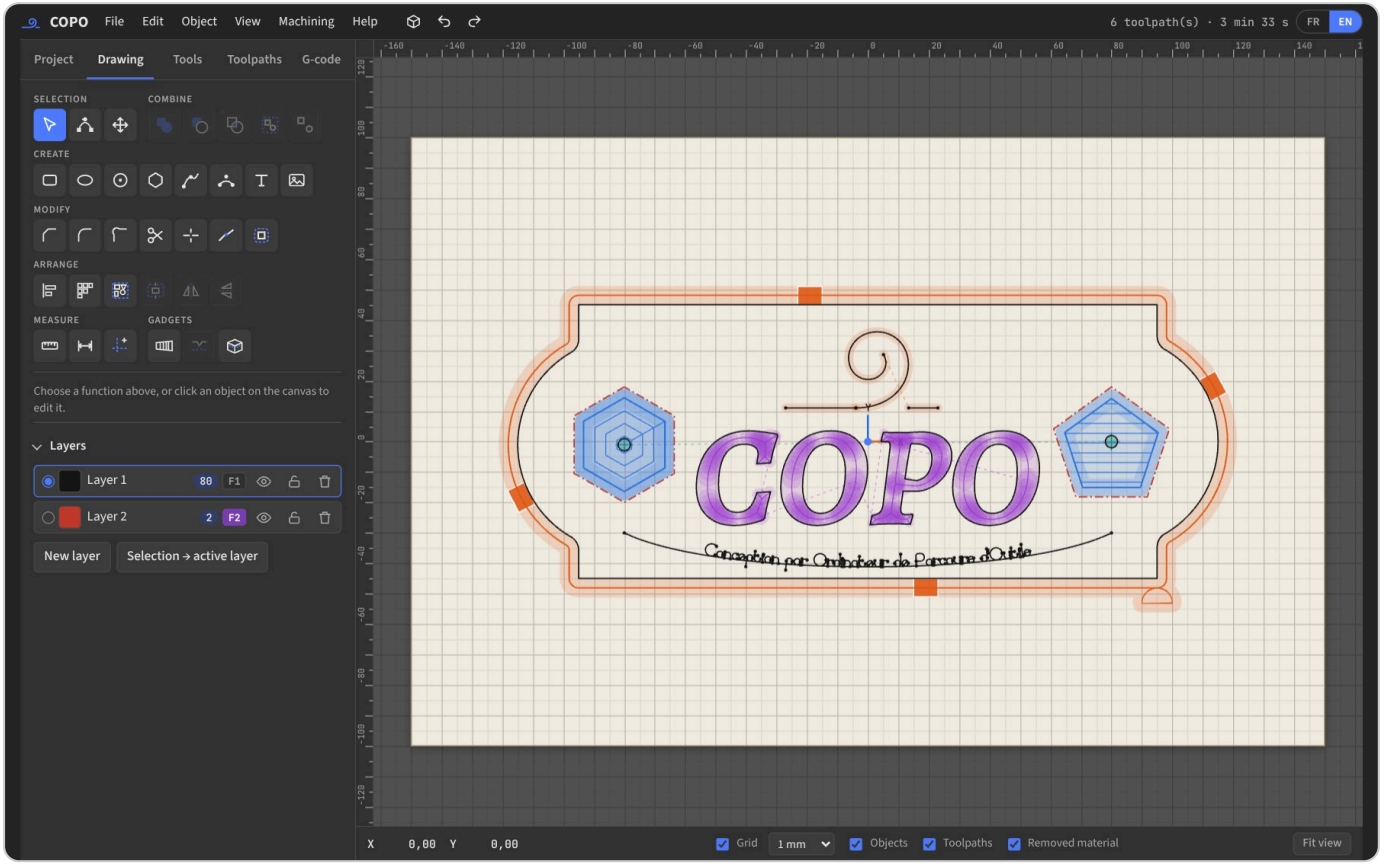
COPO draws your parts as vectors, calculates toolpaths (profile, pocket, V-carving, drilling) and writes the G-code for your machine. Everything runs in the browser, with no installation and no account.

`Version 1.65` `GPL v3` `Windows · macOS · Linux`

`Chrome · Edge · Firefox` `Français · English`

**Open the app**
https://gbulleryahen-hash.github.io/copo/

**GitHub repository (code, download)**
https://github.com/gbulleryahen-hash/copo




The sample project: COPO text V-carved, a line of single-line text on a curve, a decorative flourish with a ball nose bit, two drilled holes, a cut-out with tabs and two pockets on side 2. Toolpaths are shown in color on the stock.

# At a glance

### Vector drawing
Rectangle, ellipse, polygon, star, arc, Bézier pen, text, nodes, chamfers, fillets, booleans, layers, dimensions.

### 2D machining
Outside, inside or on-the-line profile, ring or raster pocket, rest machining, drilling.

### V-carving
Along the medial axis, like Aspire: the bit rises up to the tip of every corner.

### Tool entries
Back-and-forth ramp, continuous ramp, arc, ramped arc, helix. Allowance and finishing pass.

### Tabs
Spaced automatically or placed by hand, flat or triangular.

### Nesting
Places several parts on the sheet following their true shape, with rotations.

### Single-line fonts
About sixty included (Hershey, EMS, Relief…) for fast one-pass engraving.

### Double-sided
Layers per side, flip around X or Y, pause to turn the part over.

### 3D preview
Material removal simulation, side 1 and side 2, with animated playback.

### G-code
GRBL, manual GRBL (G53), Carbide Motion, Onefinity, LinuxCNC, Mach3/4, UCCNC, MASSO or a custom post-processor. G2/G3 arcs.

### Setup sheet
Printable workshop sheet: 2D and 3D views, origin, tools, toolpaths, speeds, checks.

### Millimetres or inches
Drawing, tools and feeds in mm or inches (fractions available), G-code in G21 or G20.

### French and English
Interface and user guide in both languages, chosen with the FR / EN buttons at the top right.

### Free software
GNU GPL v3 licence: use, modify and share COPO freely.

# Your first job in six steps

1

**Set up the project**

**Project** tab: stock size and material, X0 Y0 origin point, Z zero (top of stock or table), units and machine.

2

**Draw or import**

Draw with the icons of the Drawing tab, import an SVG, a DXF or an image (the **Import** icon in the Create group, or the File menu) or type some text.

3

**Check your tools**

Choose the material in the Project tab, then check in the **Tool library** (Toolpaths tab) the speed, feed, pass depth and stepover of each bit.

4

**Create the toolpaths**

Select shapes on the canvas, open the **Toolpaths** tab, choose Profile, Pocket, V-carving or Drilling, adjust, then click **Calculate**.

5

**Check in 3D**

**View** › **3D machining preview** shows the machined stock. Play the job back to spot a wrong depth or tool.

6

**Save the G-code**

**G-code** tab: choose the post-processor, then **Save G-code**. Always do a dry run before cutting.

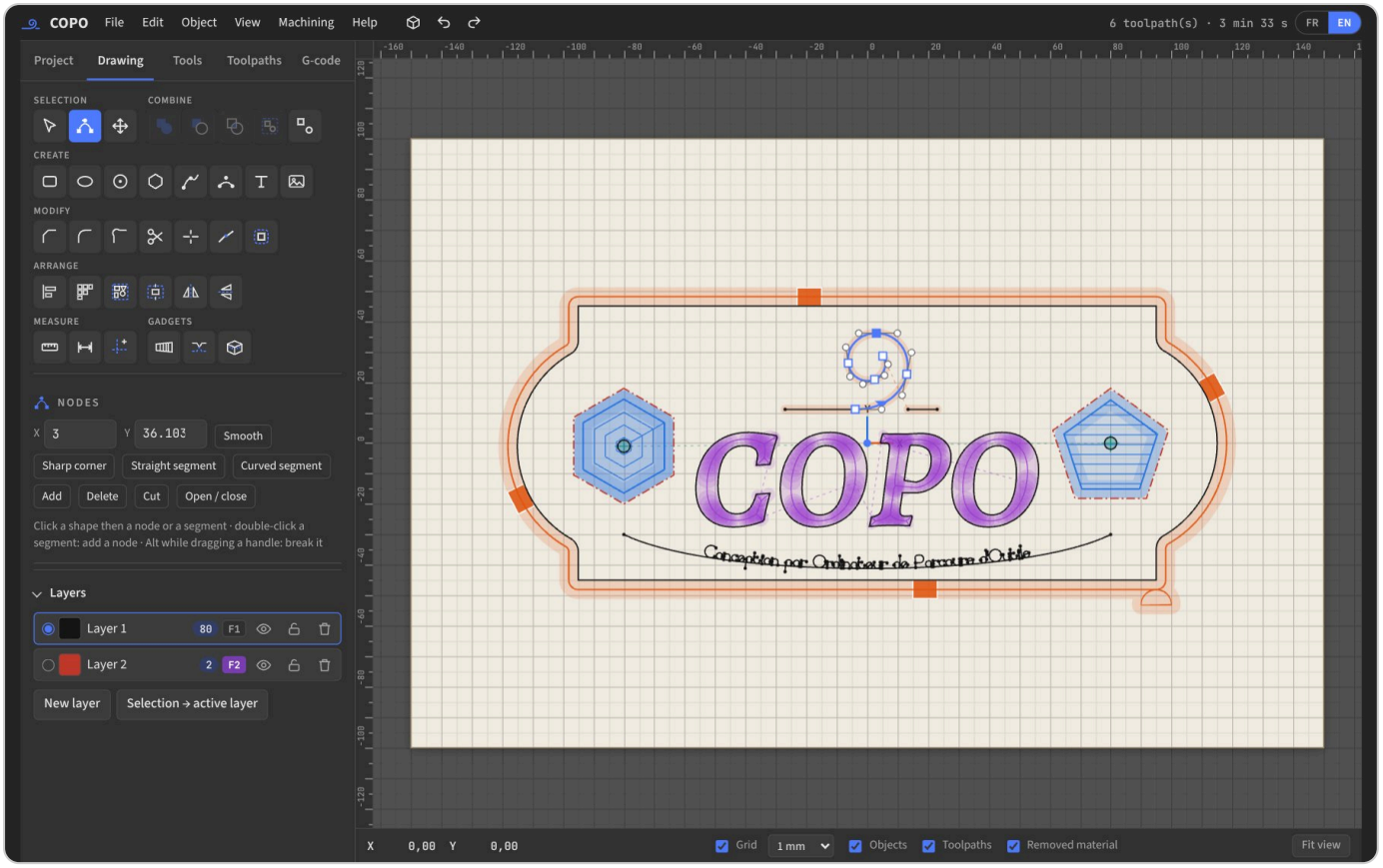
# The interface

The window reads from left to right, in the order you work.

- **Menu bar**: File, Edit, View, Drawing, Machining and Help. The Drawing menu sorts its commands into six framed categories: Groups, Combine, Vectors, Corners, Arrange and Gadgets. The Help menu opens this guide, lets you report a bug and shows the licence. The **FR / EN** buttons, at the far right of the bar, switch the interface language; the button next to them puts COPO in **full screen** (also View › Full screen; Esc to leave).
- **Drawing tab**: at the top, a grid of icons gathers every drawing function (Selection, Combine, Create, Modify, Arrange, Measure, Gadgets), each with a tooltip and a shortcut letter. Just below appears what you need: the settings of the chosen tool (with a hint and a small diagram), the form of the clicked function (Align, Array copy, Nesting, Images…), or the Selection block when an object is selected. Greyed icons need a selection first. Layers stay at the bottom. Esc closes the open function.
- **Workspace**: the stock in beige, rulers in millimetres, the grid and the origin (X and Y arrows).
- **Left panel** with four tabs: **Project**, **Drawing**, **Toolpaths** and **G-code**. They are numbered 1 to 4 in the usual order of work. In the Project and Toolpaths tabs and in the tool library, small diagrams show what each setting does (depth, allowance, cut direction, tool entry, tabs, bit shape, Z zero…) and update with your choices.
- **Status bar**, at the bottom: cursor coordinates relative to the origin, then two groups of on/off buttons. *Snap* (magnet): grid and its spacing, objects, guides, and the settings button that opens Snap settings. *Show* (eye): grid, toolpaths, removed material. A button that is on is highlighted in blue.

A floating bar appears under the selection: **Duplicate**, **Group**, **Weld**, **Machine…** and **Delete**. The project is saved automatically in the browser; **File** › **Save project** writes a copy to a file.

# Drawing




The Nodes tool on the decorative flourish: the options bar turns a node into a sharp or smooth corner, a segment into a line or a curve, and adds, deletes or cuts a node.

## Shapes and paths

- **Rectangle** `R`, with sharp, rounded or chamfered corners. **Ellipse** `E` (Shift: circle) and **Circle from center** `C`.
- **Polygon and star** `O`: number of sides or points, inner radius.
- **Pen** `P`: click for a corner point, click-drag for a curve point, click the first point to close, double-click to finish. The tool settings take absolute, relative or polar coordinates: type the values then press `Enter`.
- **3-point arc** `A`: start, end, then the bulge.
- **Nodes** `N`: move nodes and handles, double-click a segment to add a node, `Alt` while dragging a handle to break it. Three node types: **Smooth** (circle, aligned handles), **Cusp** (diamond, two independent handles for a corner between two curves) and **Sharp corner** (square, no handles). Several nodes: `Shift`+click to add or remove one, drag a box in empty space to take them all at once (`Shift` to add), `Ctrl+A` for all. Selected nodes move together with the mouse or the **arrow keys** (1 mm, `Shift` 10 mm, `Alt` 0.1 mm); Smooth, Sharp corner and Delete apply to all of them. A selected **segment** (click it, it turns red) is deleted with **Delete** or the `Del` key: a closed outline

becomes an open line, an open line is split in two. To switch to another shape, click its outline.

## Editing

- **Chamfer** `K` and **Fillet** `F` on one corner, or on every sharp corner from the **Corners** block.
- **Trim** `X` deletes the part of a vector between two intersections; **Cut** `U` splits a vector where you click.
- **Weld, Subtract, Intersect** for boolean operations; **Join** and **Close** for open vectors, gaps being bridged by a straight line or by a smooth curve that continues both vectors; **Offset contour**: on a closed vector outside, inside or both sides; on an open vector to the right or left of the path, or all around ("Both" creates a closed outline). Sharp or rounded corners, and an option to delete the original.
- **Smooth (fit curves)** (Modify): replaces vectors made of many small segments (tracing, DXF, import) with *arcs*, *Bézier curves* or *straight lines* that stay within the tolerance of the original. "Keep sharp corners" keeps as sharp corners those whose change of direction exceeds the max angle (60° by default). A dashed orange preview and the node count before/after are shown before applying; untick "Replace" to keep the originals. *Arcs* or *Bézier*? In COPO every curve (circles, arcs, ellipses, letters) is stored as Bézier curves, as in an SVG file. The *Arcs* mode finds the portions that follow a true circle (constant centre and radius) and draws them with Béziers fitted to that arc, at most one per quarter circle: the gap from a perfect arc is about 0.03% of the radius (under a hundredth of a millimetre for a 30 mm radius). Use it for mechanical shapes (templates, fillets, holes): the radii are truly constant and give clean G2/G3 arcs. The *Bézier curves* mode gives free-form curves, often with fewer nodes on organic shapes (traced drawings, scrolls), but a "round" part is not necessarily an exact circle.
- **Align and distribute**: choose the reference in "Align to". *The selection*: a single object aligns to the stock, several align to each other. *The first selected object*: it stays put and the others align to it; Distribute places them after it, in the order you selected them, with the given spacing. *The stock*: aligns to the stock edges and distributes across its full width or height.
- **Rotation** (Arrange): an orange pivot appears at the centre of the selection; drag it on the plan (it snaps to objects, guides and the grid), pick one of the nine points, or type its coordinates. *Relative* rotation (the selection turns by the given angle, positive counter-clockwise) or *absolute* (the selection takes the given angle; COPO remembers earlier rotations). ↺ 90° and ↻ 90° buttons for quarter turns.

- **Handles**: click an already selected object a second time. Squares resize (corners keep proportions when “Keep proportions” is ticked, Shift for the opposite; middles resize one axis), circles rotate around the pivot (Shift: 15° steps), and the pivot can be moved. Click again to hide them.
- **Right-click and double-click**: a single right-click on the plan finishes the current line (polyline), cancels a shape in progress, goes back to the selection tool and deselects; with the selection tool, it hides the handles and then deselects. Double-clicking an object, from the selection tool or a shape tool (rectangle, ellipse, circle, polygon, arc, dimension, measure), selects it with its handles; another double-click opens node editing (Esc or right-click to leave).
- **Array copy** in a grid or a circle, **mirrors**, **rotation**, **stacking order**.
- **Exact position and size**: **Selection** block, with a reference point chosen among nine (corners, middles, center).

## Snapping, guides and dimensions

**Edit** › **Snapping…** opens the **Snap settings** window: snap to grid and its spacing, **guides**, stock corners, midpoints and centre, snap radius (small to large), and object snapping with its five points (ends and nodes, midpoints, intersections, centres, tangents). These settings are remembered. Drag from a ruler to create a guide: drawn points snap to it, and when you move a selection its edges and centre line up with the guides (and with the stock if that box is ticked). An orange symbol at the cursor shows the snap found. **View** › **Show grid** shows or hides the canvas grid. The **Measure** tool `M` gives a distance and the **Dimension** tool `D` places a dimension that stays on the canvas. An end snapped to a node, a segment midpoint or the centre of an arc or circle (solid dot at the end of the extension line) stays attached to the object: move it, resize it or edit its nodes and the dimension follows, with its value updated. An end placed elsewhere (grid, intersection, free point) stays fixed.

## Layers, images and files

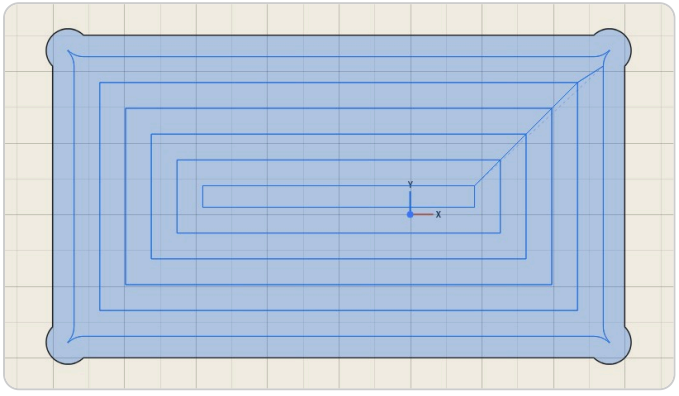
- **Layers**: stroke color, visibility, lock and machined side (F1 top, F2 bottom).
- **Reference images** (**Import** icon › Image, **File** › **Insert an image…** or drag and drop): move, resize and rotate them with the handles to trace over them. They are never machined.
- **Vectorize an image** (Drawing panel › Images, image selected): COPO traces the outlines of the image. Set the *threshold* (anything darker becomes the shape; “Auto threshold” works it out for you) and the *smoothing* (softer outlines, small details removed), tick *Invert* to outline light areas. The preview updates live; **Create outlines** adds closed, grouped shapes where the image

is, ready for a pocket, a profile or V-carving. A sharp, high-contrast image (logo, line drawing, silhouette) gives the best result.

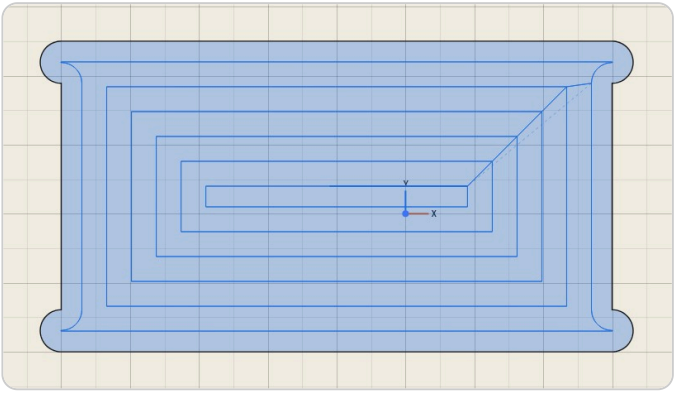
- **Import**: the **Import** icon in the Create group opens a panel with two choices, *Vector* (SVG, DXF) and *Image* (template to trace over or vectorize).
- **SVG**: import in millimetres with curves kept; export the whole drawing or the selection.
- **DXF** (text/ASCII, any version): lines, polylines with arcs, circles, arcs, ellipses, splines and blocks. The file's unit is respected (mm, cm, inches…); a file without a unit is read as millimetres. Touching segments are joined into outlines, closed when possible, and each DXF layer becomes a COPO layer. Text, dimensions, hatches and 3D objects are skipped (the message says which).

## Corners for tenons and mortises: dogbone and T-bone

A round bit leaves a fillet of its radius in corners, so a square tenon will not fit its mortise. The relief is drawn on the shape, corner by corner, so it only goes where it is needed. Take the **Relief** tool `B` (Modify group), choose *Dogbone* (the bit moves diagonally to touch the sharp corner) or *T-bone* (the relief extends the longer side, less visible) and the **bit diameter**, then click near the corners to relieve. The same settings are in the **Corners** block of the Drawing panel. Corners are picked one by one, as for chamfers and fillets. Relief only works between two straight segments; its diameter is 0.1 mm larger than the bit so it fits without forcing. Then machine the shape as usual (pocket, inside or outside profile) with that bit.

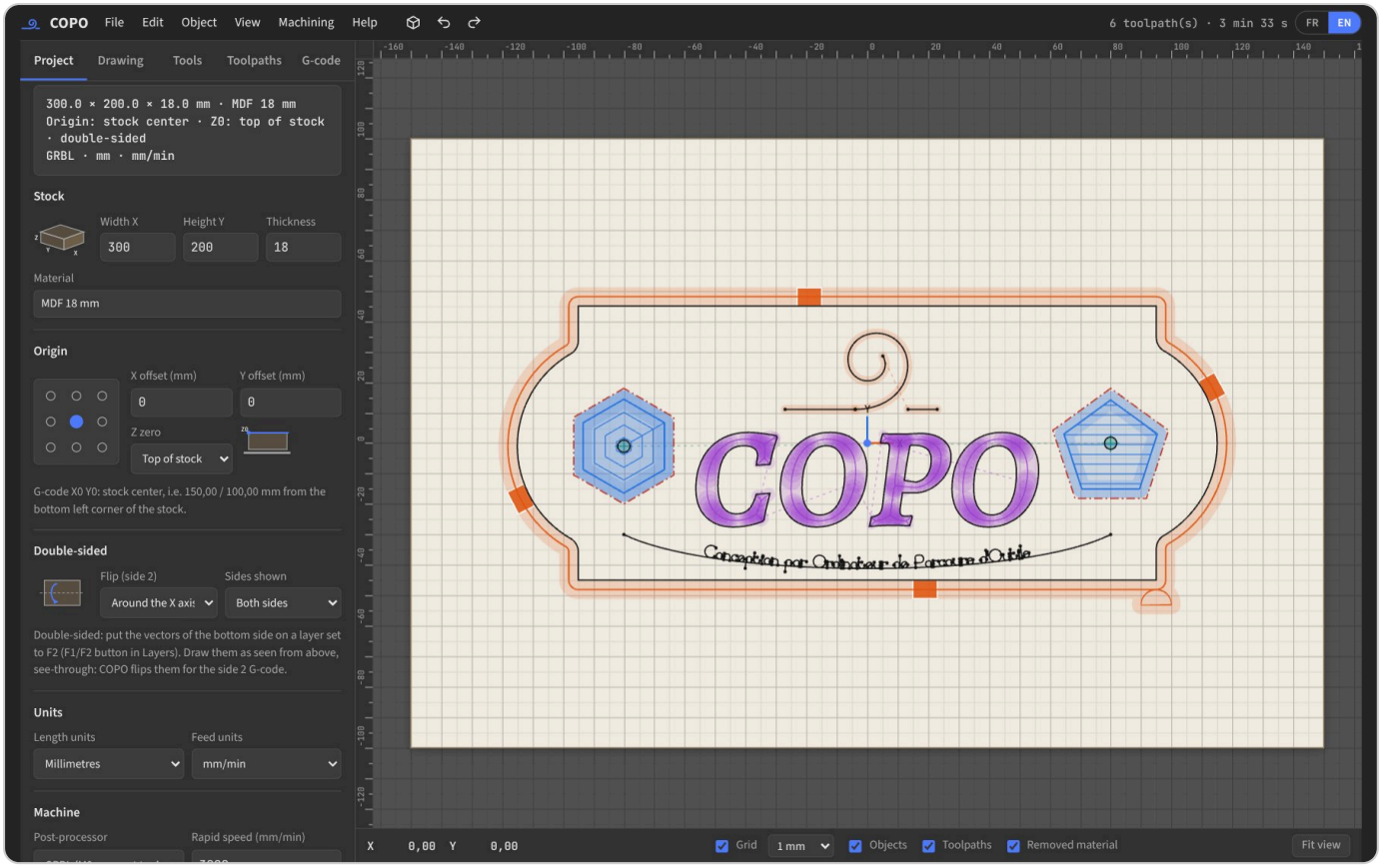


**Dogbone**: an arc the size of the bit, diagonally in each corner.



**T-bone**: the arc extends the longer side.

# Project tab




The Project tab gathers everything you set at the start of a job. The summary at the top lets you check it all at a glance.

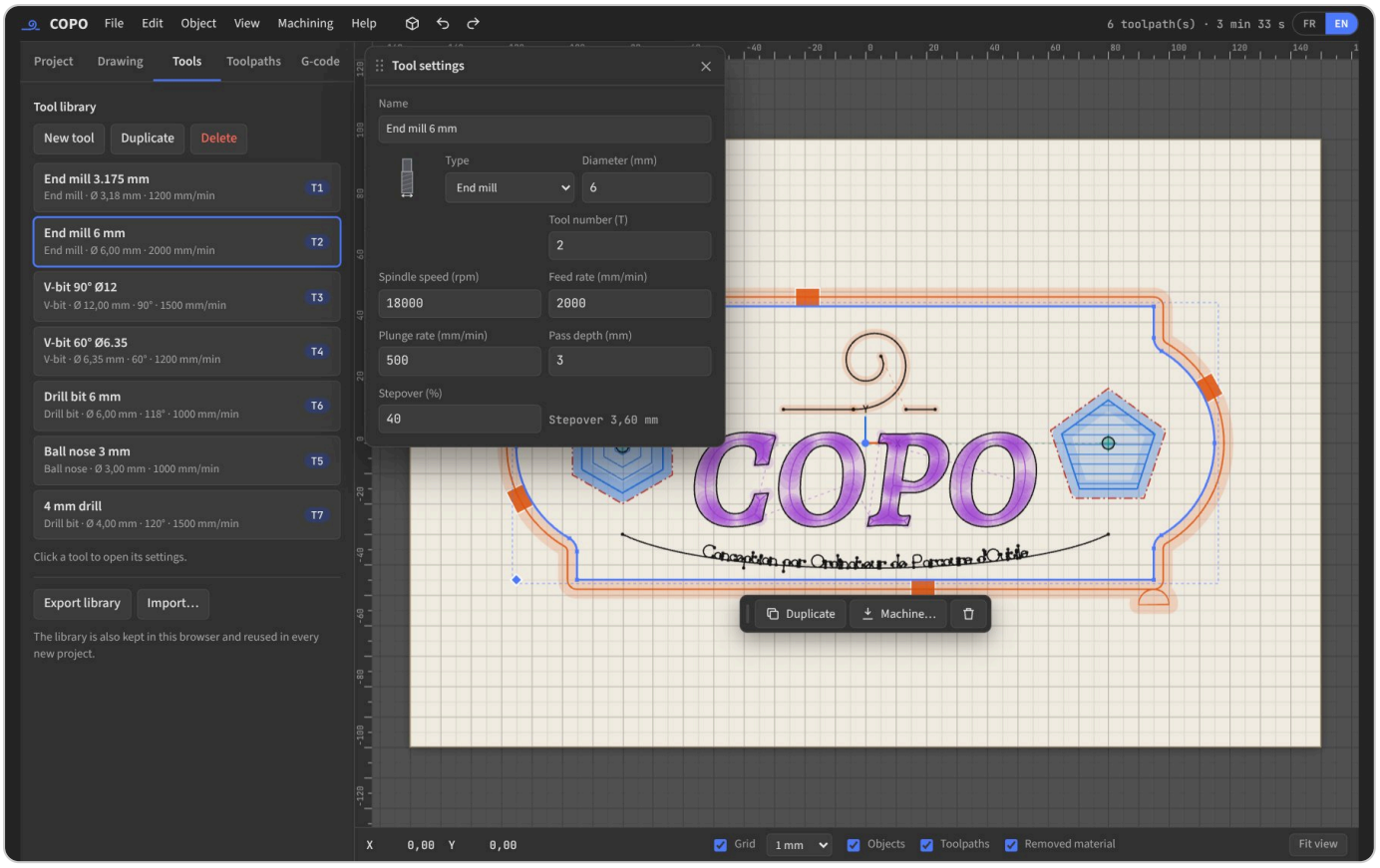
The **Project** tab, first in the left panel, opens by itself when you create a new project. It contains:

- **Stock**: X width, Y height, thickness and material (used by the setup sheet);
- **Origin**: the X0 Y0 point on the nine-point grid, an X/Y offset to place it outside the stock, and Z zero on the top of the stock or on the table;
- **Double-sided**: the flip axis and the sides shown;
- **Units**: millimetres or inches, feed unit;
- **Machine**: post-processor, rapid speed, **maximum cutting feed** (2000 mm/min by default: tool feeds are capped at this value), safe and clearance Z, G2/G3 arcs and comments.

The origin point must match the point where you set X0 Y0 on the machine. Stock, origin and material are saved with the project; units and machine settings are remembered in the browser.

> Example: with the origin at the bottom left corner, a square whose corner sits at X0 Y0 on the canvas is cut from the bottom left corner of your sheet. The line under the grid tells you where the G-code X0 Y0 falls.

# Tool library




The tool library: the material at the top, tools sorted by unit system and type on the left, the tool's settings for that material on the right.

The library opens from the **Tool library** card in the Tools block of the Toolpaths tab (it shows the project material), from **Machining › Tool library…**, and from the **Tool** button of a toolpath window. Inspired by Aspire's, it comes filled: metric (mm) and imperial (inch) tools, sorted into end mills, ball nose, V-bits, engraving bits and drills, with settings for seven materials: **softwood, hardwood, MDF, acrylic, PVC, aluminium and brass**. The first four come from an Aspire library tuned for a WorkBee-type machine; settings missing there (large bits), PVC (like acrylic), aluminium and brass are **estimated** for a hobby CNC, at the low end of the ranges recommended for light machines (Shapeoko, X-Carve, 3018, Stepcraft; cross-checked with Carbide 3D and Sienci): in aluminium about 0.015 mm per tooth at Ø 3 and 0.025 mm at Ø 6, brass 20% lower. Estimated values are flagged by an orange banner: do a test cut and correct them. These values are

deliberately cautious; makers' charts, written for rigid machines, go 3 to 5 times further.

- **Material** (top): this is the project material, also set in the Project tab (*Tool material*). Changing it makes every project tool take that material's settings and recomputes the toolpaths. The **+** button adds a material.
- **Tree** (left): click a folder to open it, double-click to rename it; the search box filters by name; “used” marks the tools of the project's toolpaths. At the bottom: new tool, new folder, duplicate, delete and back to the default library (two clicks). The **Import…** and **Export…** buttons are at the top right.
- **Settings** (right): geometry (name, type, diameter, angle, engraving tip, number of flutes, T number, notes), then cutting settings for the chosen material. The name updates itself as long as you have not edited it.

| SETTING | PURPOSE |
| --- | --- |
| Type | End mill, ball nose, V-bit (V angle), tapered flat-tip engraving bit (angle and tip diameter), **tapered ball-nose** or drill (point angle, usually 118°). |
| Tapered ball-nose bit | Defined as on its box: **shank diameter D**, **tip radius R** and **cutting length L**. COPO derives the side angle, that of the cone tangent to the ball that reaches diameter D at height L: for D 6, R 1, L 20, about 5.99° (11.99° included). Engraving and tapered bits are used like V-bits of that angle (V-carving, on-the-line profile); the rounded or flat tip is not yet taken into account in the depth. |
| Diameter | Used to calculate offsets and pockets. |
| Tool number (T) | Written in the G-code of tool-change post-processors ( `T2 M6` , `M6 T2` ). |
| Spindle speed | In rpm, written with `M3` . |
| Feed, plunge rate | In mm/min, for cutting and for plunging or ramping. |
| Pass depth | Maximum depth per pass; it fills the toolpath's pass depth when you choose the tool. |
| Stepover | Distance between two pocket passes, in mm or % of the diameter (as in Aspire). |
| Number of flutes | Number of cutting edges of the bit. |
| Feed per tooth | Calculated: feed ÷ (spindle speed × number of flutes). Type a value to recalculate the feed. The comment below, on an orange background when conditions are not optimal, says whether it is low (the bit rubs and heats up) or high (risk of breakage), against the hobby-CNC ranges (wood and MDF, plastics, aluminium, brass) by diameter, and warns when the feed exceeds the machine's maximum feed; the bit maker's data come first. For a drill, COPO gives the feed per revolution. |

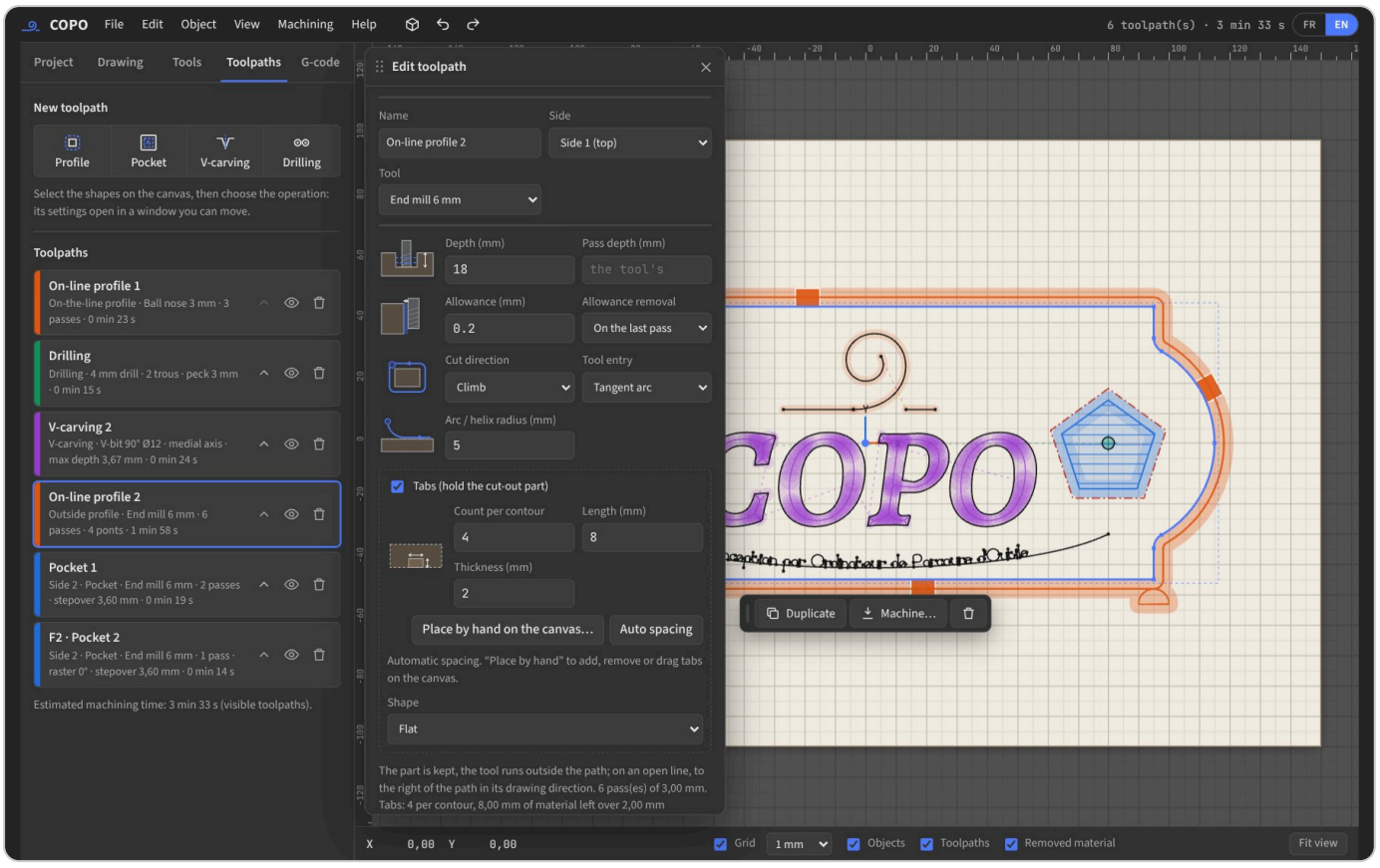
## Choosing a toolpath's tool

In a toolpath window, the **Tool** button shows the chosen tool; click it to open the library, select the tool then **Use this tool** (or double-click). Tools that do not suit the toolpath type are greyed out. The toolpath takes the tool's settings for the project material, and its pass depth. Editing a tool in the library updates the toolpaths that use it.

## Importing an Aspire library

The import button reads Aspire, VCarve and Cut2D libraries directly ( `.vtdb` file, for example `C:\ProgramData\Vectric\Aspire\V12.0\ToolDatabase\tools.vtdb` , or an export from Aspire's library). Tools, their folders and their per-material settings are added in an “Aspire import” folder, inches converted to millimetres; if the file holds several machines, COPO asks which one to use. Missing materials (PVC, aluminium, brass) are estimated. Form tools and diamond drag bits, which COPO cannot machine yet, are skipped. Import also accepts a COPO library exported as `.json` .

# Toolpaths




Click a toolpath in the list to edit it: its path is highlighted on the canvas. The text under the form sums up what the machine will do.

At the top of the tab, the four buttons **Profile**, **Pocket**, **V-carving** and **Drilling** create a toolpath: select the shapes on the canvas (even with the window open), then click the operation. Its settings open in a floating window you can move by its title; COPO remembers its position. Clicking a toolpath in the list opens the same window to edit it.

While the toolpath window is open, the rest of the drawing and the other toolpaths are **dimmed**: only the selected vectors and the current toolpath stand out. Everything returns to normal when the window closes. At the bottom of the window, a **status line** says what will be machined, for example “3 vectors selected (2 closed, 1 open)”; it turns red when nothing is selected, and warns when a pocket, V-carving or drilling will ignore open vectors.

## The four types

- **Profile**: *Outside / right* (the part is kept), *Inside / left* (the hole is cleared) or *On the line*. On an open line, Outside runs the tool to the right of the path and

Inside to the left, in its drawing direction (Drawing › Reverse direction to change it); “On the line” follows the stroke or runs left or right according to the “Open lines” setting. On an open line offset to the right or left, ramp, tangent arc and ramped arc entries apply too: the arc enters and leaves on the waste side.

- **Pocket**: in *rings* parallel to the contour or as a *raster* at the chosen angle. *Rest machining* uses a smaller bit for the corners a larger bit could not reach: keep the same depth and place this toolpath after the original pocket.
- **V-carving**: the V-bit follows the medial axis of the path; it goes deeper where the stroke is wide and rises up to the tip of every corner. A maximum depth gives a flat bottom.
- **Drilling**: one hole at the center of each closed shape, with pecking and a dwell at the bottom. Circles larger than the bit are milled helically to the drawn diameter. With a drill bit, each hole is drilled at the bit diameter; COPO shows the tip length to add for a through hole.

## Passes, allowance and cut direction

**Start depth (D)** and **Cut depth (C)**, as in Aspire: machining starts at depth D below the surface (for example at the bottom of a pocket already cut) and goes down by C from there, so the bottom is at D + C. Leave D at 0 to start from the surface. In V-carving, the pattern is computed as if the surface were at D, which is what inlays use. The small diagram next to the fields shows what D and C are.

The cut depth is split into passes of the chosen depth. An **allowance** leaves a little material on the walls while roughing; it is removed on the last pass or by a full-depth finishing pass. The cut direction is *climb* or *conventional*.

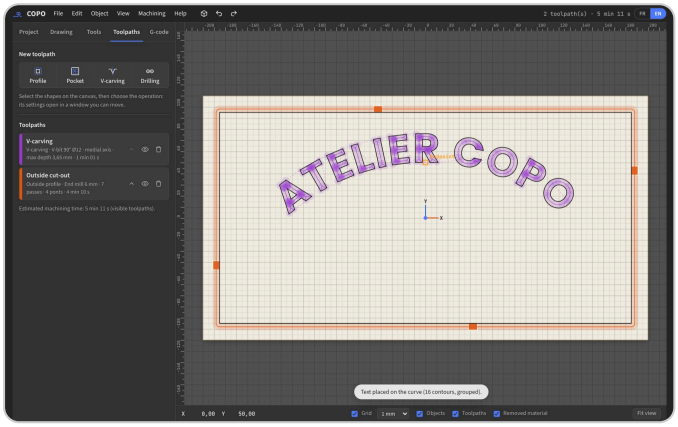
### Tool entry

| ENTRY | BEHAVIOUR |
|---|---|
| Back-and-forth ramp | The tool goes down while moving, returns to the start at full depth, then goes around. Each pass starts and ends at the same point. Default setting. |
| Continuous ramp | The tool goes down while moving and continues around; the ramp portion is recut at the end. |
| Tangent arc, ramped arc | Arc lead-in and lead-out on the contour, to avoid a mark on the finished wall. |
| Helix | The tool spirals down, ideal for pockets and holes. |
| Straight plunge | Vertical plunge, for center-cutting bits or V-carving. |

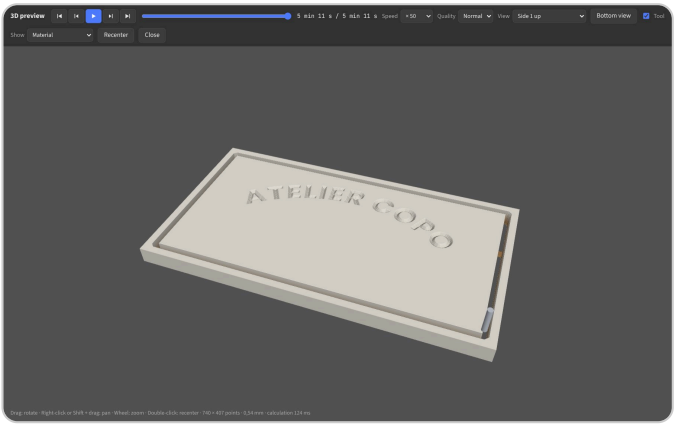
### Tabs

Tabs hold a cut-out part until the end. Set their count per contour, length and thickness, flat or triangular (easier to break). **Place by hand on the canvas…** lets you add a tab with a click on the toolpath, drag it or remove it.

# Text and V-carving




Text placed on an arc, then V-carved.




The same job in the 3D preview.

Choose a font in the **Text** block list: installed fonts (Chrome and Edge) or your fonts folder (**File** › **Open a fonts folder…**), with a preview. Two boxes at the top of the list, *COPO fonts* and *Computer fonts*, shorten the list; the choice is remembered. Take the **Text** tool T and click on the canvas. With **On a curve** placement, click a contour: the text centers on it, above or below, with the

spacing you want. The text becomes grouped vectors, ready for V-carving, pocketing or profiling.

### Single-line fonts

COPO comes with about sixty **single-line fonts** (Hershey, EMS, Cutlings, Relief SingleLine, Routed Gothic…), stored in the `fonts/trait-unique` folder and listed first in the font list, with a preview. Each letter is a single stroke: the text is engraved in one pass with a **Profile On the line** toolpath, with a V-bit or a small end mill. It is the fastest way to mark part numbers, plates and assembly marks. When the selection holds only open strokes, **Machine…** offers this profile directly. The `fonts/apercu-polices.html` catalogue (link “See the single-line font catalogue” in the Text block) shows them all with a preview of your text. You can also add your own SVG fonts with **Add fonts…**.

### Decorative fonts

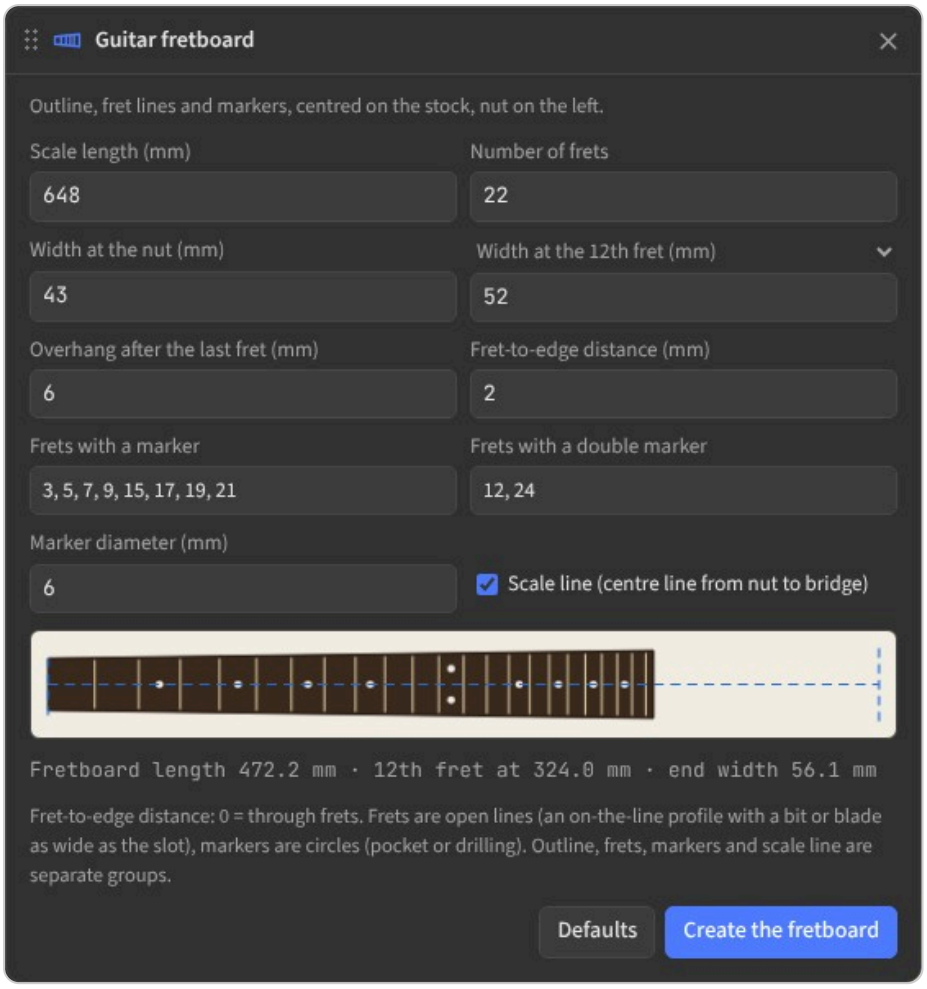
Nine **decorative font** families are also bundled, in the `fonts/decoratives` folder, and listed under *Decorative fonts (bundled with COPO)*: **URW Chancery** (chancery italic), **Alegreya** (roman and italic), **Cinzel Decorative** (ornate capitals, 3 weights), **Pirata One** (blackletter), **Chicle**, **Comic Neue** (6 styles), **Protest Riot**, **Great Vibes** (script) and **Smokum** (western). These are outline fonts: machine them with V-carving, a pocket or a profile. For thin scripts such as Great Vibes, use a large enough text height (20 mm or more) so the bit fits in the hairlines. The *COPO fonts* box hides them together with the single-line fonts. Licences: SIL Open Font License for most, Apache 2.0 for Smokum, AGPL 3 with font exception for URW Chancery (see `fonts/decoratives/LICENCES.txt`).

## Lutherie: fretboard and inlay

### Guitar fretboard

**Gadgets › Guitar fretboard** (or Drawing › Guitar fretboard…) draws the fretboard, nut on the left, centred on the stock. Set the **scale length**, the **number of frets**, the **width at the nut** and the second width, *at the 12th fret* or *at the end of the scale*, the overhang after the last fret, the **frets with a marker** and **a double marker** (lists of numbers) and the marker diameter. The **scale line** (ticked by default) draws the centre line from the nut to the bridge, with a mark at each end, to place the bridge. With the **Zero fret** box, a fret 0 is drawn at the start of the scale (with the same edge distance as the others) and the board starts before it, at the distance set in *Board start → zero fret*; the “nut” width is then the zero-fret width. The **fret-to-edge distance** stops the fret lines short of the board edge (0 =

through frets). The preview updates live. The window opens next to the left panel, like the toolpath settings, and moves by its title.




The **Guitar fretboard** window with its preview: board, frets, markers and the dashed scale line.

### Fret positions

COPO uses the exact equal-temperament formula, the one used by reference fret calculators:

> $\mathbf{d_n = L - L / 2^{n/12}}$ i.e. $d_n = L \times (1 - 2^{-n/12})$

where $L$ is the scale length, $n$ the fret number and $d_n$ the distance from the nut to the fret. Each fret shortens the string by a semitone, a factor of $2^{1/12} \approx 1.05946$; the $12^{th}$ fret falls exactly at half the scale. The “rule of 17.817” gives the same result; the old, less accurate rule of 18 is not used. No compensation is applied to the frets: intonation is set at the bridge, moved back a few millimetres from the scale line.

| FRET | 1 | 5 | 7 | 12 | |
|---|---|---|---|---|---|
| Distance from the nut (648 mm scale) | 36.37 mm | 163.51 mm | 215.64 mm | 324.00 mm | 466.1 |

The result is separate groups: the scale line (not to be machined), the outline, the fret lines (open lines, to cut with an **On the line** profile and a bit or blade as wide as the slot) and the markers (circles, pocketed or drilled).

### V-carve inlay

Select the design (closed shapes), then **Drawing** › **V-carve inlay…**. Choose the V-bit, the **female pocket depth**, the **glue gap** and the **male cut depth**. COPO creates:

- a *female* V-carving of the design in your part;
- the *male* plug: the mirrored design inside a frame, on the “Inlay · male” layer, with a V-carving that starts at the pocket depth minus the gap. Its toolpath is created *hidden*: to cut it from the inlay wood, show it and hide the female one before saving the G-code.

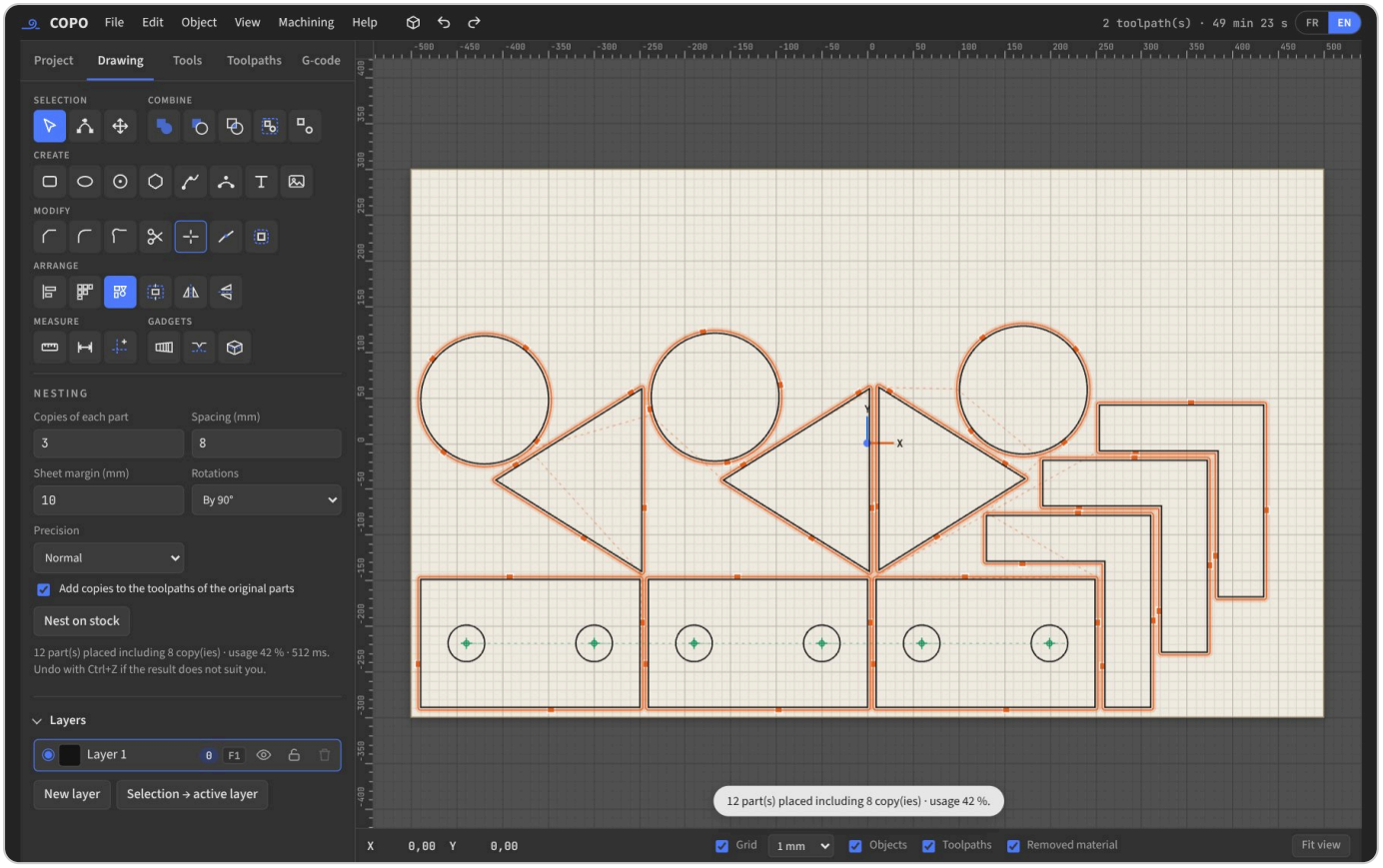
Flip the plug, glue it into the female pocket, then plane or sand off the excess. The V-carving **Start depth** setting is also available by hand in the toolpath form.

## Finger-joint box

In the **Gadgets** group of the Drawing tab (or **Drawing** › **Finger-joint box…**), this module draws a box assembled with straight finger joints, like Vectric's “Box Creator”. Set the **outer dimensions** (width X, depth Y, height Z), the **material thickness** (the stock's by default), the **finger width** (their number, always odd, follows), the **joint allowance** (0.1 to 0.2 mm in wood), the **cutting bit**, the **top joint** (*finger-jointed* for a closed box, the default, or *laid flat* for a removable lid) and the **faces to machine** (Top, Bottom, Sides, Ends). Fingers are spread evenly between the corners: the corner zone, as thick as the material, only exists where another panel joins, so an open box keeps the same pitch up to the top. Fingers may be narrower than the thickness; COPO warns when the slots become narrower than the bit. The *Dogbone reliefs* option adds an arc the size of the bit at the bottom of each slot so square fingers fit.

The preview shows the panels laid flat. **Create the box** places them on the stock, spaced for the bit: cut them with an outside profile and tabs. The panel volumes fit exactly: no overlap or gap at the corners.

# Nesting




Four parts in three copies on a 1000 × 600 mm sheet. The L shapes interlock and the copies join the toolpaths of the original parts.

**Drawing › Nesting…** opens the block in the Drawing tab. Select the parts (with no selection, the whole drawing is used) and set the number of copies, the spacing, the sheet margin, the allowed rotations and the precision. COPO places the parts following their true shape, largest first, as far down and left as possible. A group or a text stays in one block. Parts that do not fit stay selected, and the sheet usage is shown.

> For cut-outs, use a spacing at least equal to the bit diameter. `Ctrl` + `Z` undoes the nesting if you don't like the result.

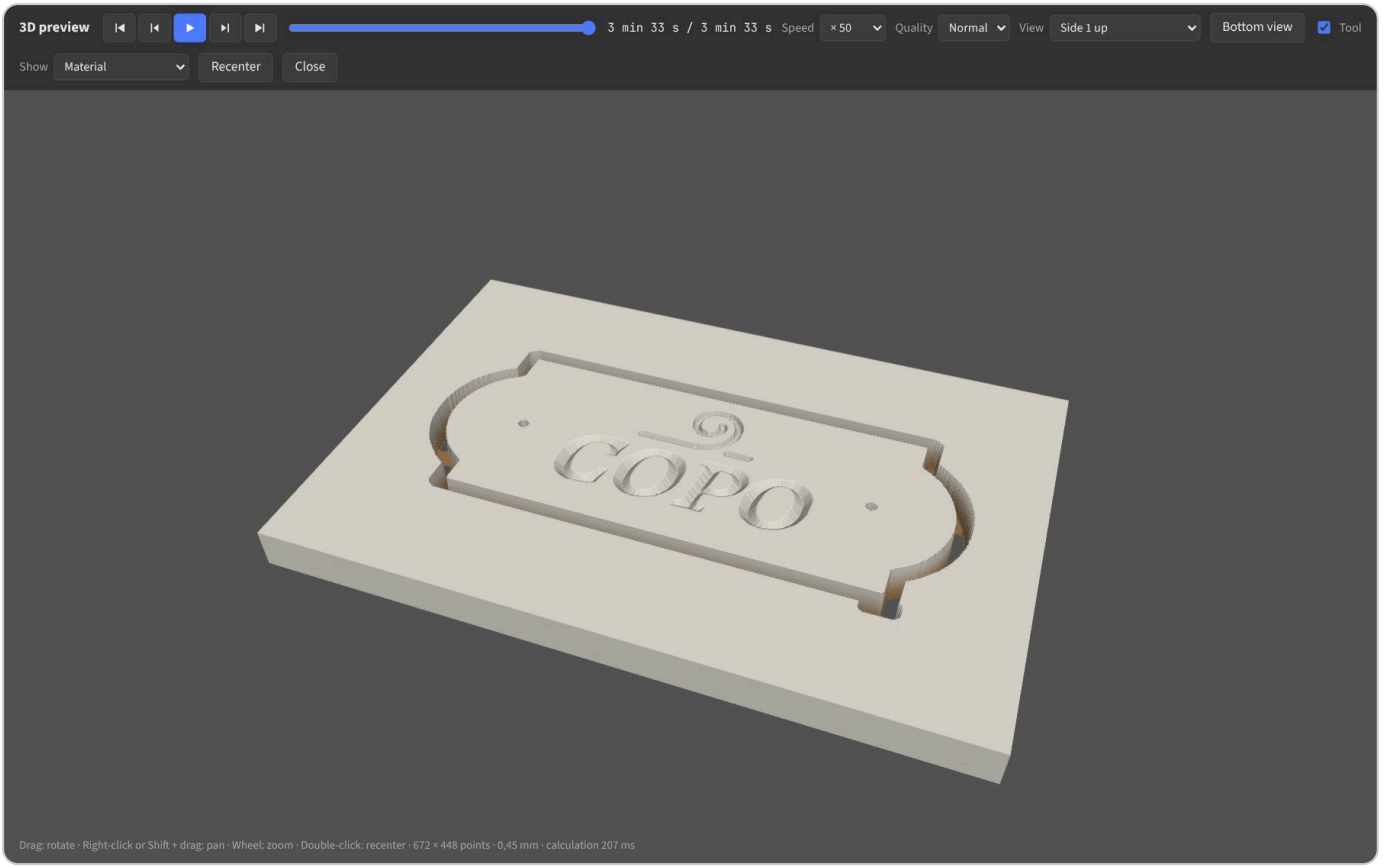
# Double-sided machining

1. Put the vectors of the bottom side on a layer set to **F2** (F1/F2 button in the layer list). Draw them as seen from above, see-through.
2. Choose the flip: around the X axis (top ↔ bottom) or the Y axis (left ↔ right).

3. Side 2 toolpaths are calculated on the flipped geometry; the cut direction is kept.
4. In the G-code, side 1 is followed by an `M0` pause: turn the part over, set X0 Y0 Z0 again at the same point and resume.

The 3D preview shows both sides; the **Side 2 up** option shows the part flipped.

# 3D preview



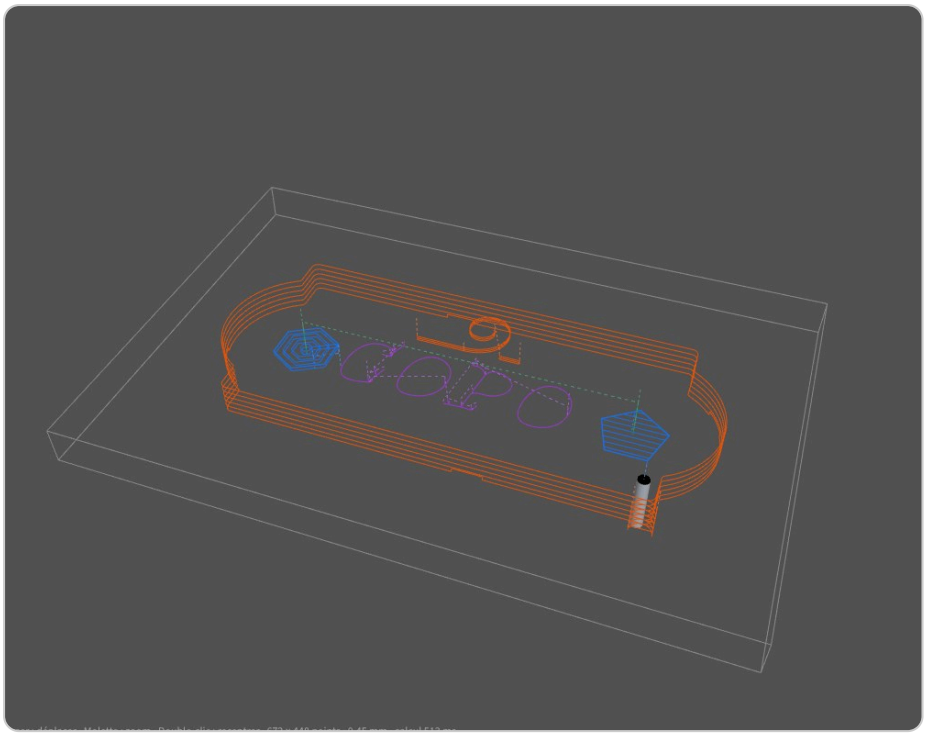

Drag to rotate, right-click or Shift to pan, wheel to zoom, double-click to recenter.

The preview simulates material removal through the real stock thickness, through-holes included. The playback buttons replay the machining: **back to start**, **step back**, **play / pause**, **step forward** and **go to end**; one step is one tool move. Keyboard: left and right arrows, Home, End and Space. The slider jumps to any moment.

## Toolpaths in 3D

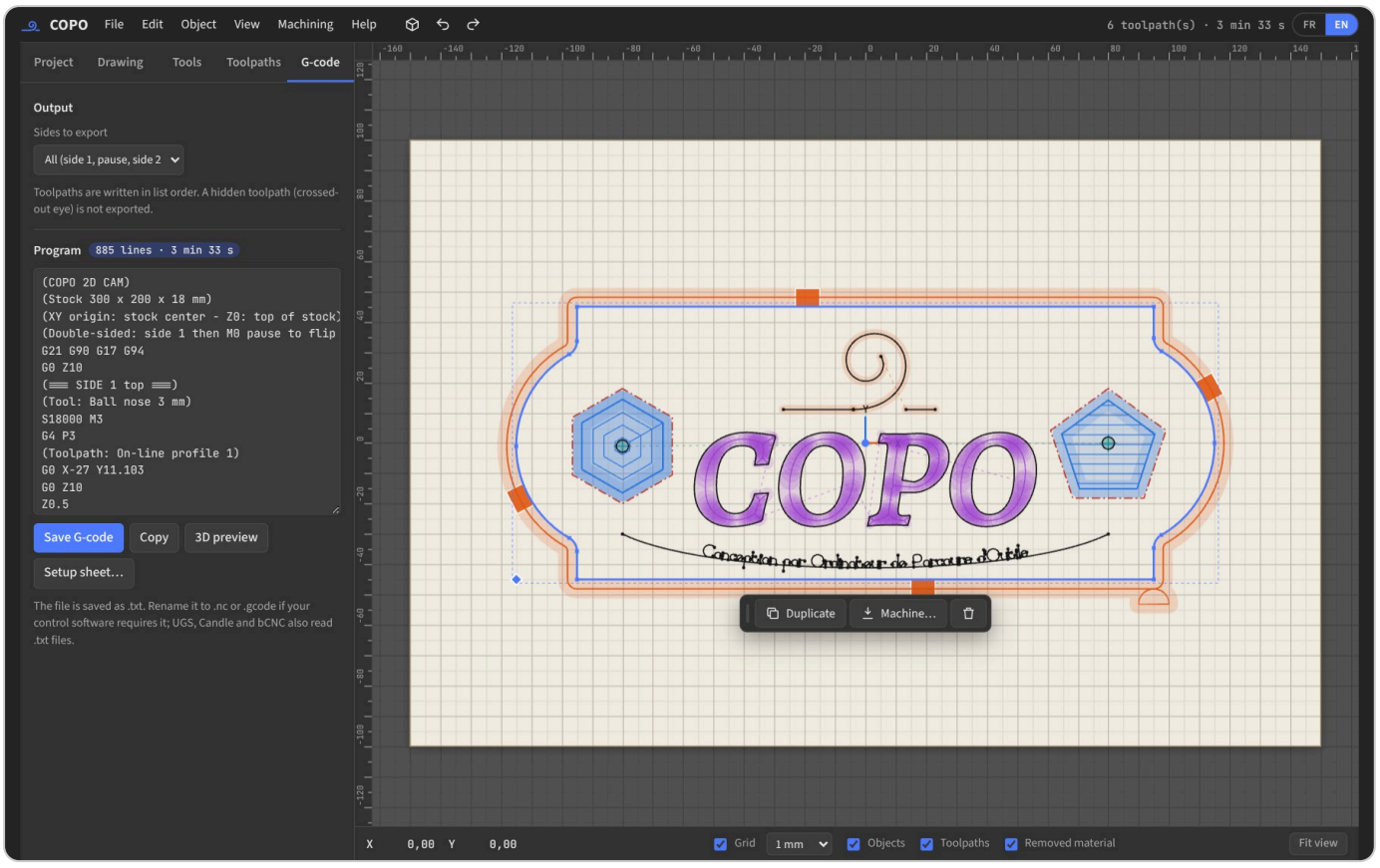
The **Show** list in the preview bar offers *Material*, *Material + toolpaths* or *Toolpaths only*. The **Drawing** box shows or hides the drawing's vectors, in their layer colour, on the face turned upwards: face 1 vectors in the Face 1 view, face 2 vectors in the Face 2 view. Toolpaths are drawn as lines in space, in the colour of their type;

rapid moves (G0) are dashed grey. During playback they are drawn as the tool moves. With a toolpath selected in the list, the **Selected toolpath only** box shows just that one. Great for checking passes, ramps, helixes and clearance heights before cutting.



*Toolpaths only* mode: every cut-out pass is visible, rapids are dashed.

# G-code




Toolpaths are written in list order; a hidden toolpath is not exported.

| POST-PROCESSOR | TOOL CHANGE |
|---|---|
| GRBL | `M0` pause between two tools. |
| GRBL manual | Before each tool, the first one included: retract with `G53 G0 Z-10` (adjustable) and `M0` pause. Also a `G53` retract at the end of the program. |
| Carbide Motion, Onefinity | `M6 T…` : the control software asks you to change the tool. |
| LinuxCNC, Mach3/4, UCCNC, MASSO | `T… M6` with the tool number, `G40 G49 G80` header, arcs with `G91.1` . `.ngc` file (LinuxCNC), `.tap` (Mach, UCCNC) or `.nc` . |
| Custom | You write the header, tool change and program end, with variables: `{UNITS}` , `{SAFEZ}` , `{TCZ}` , `{T}` , `{S}` , `{D}` , `{TOOL}` . File extension and `G91.1` are up to you. |

```
(Load tool: End mill 6 mm D6 - set Z zero again then resume)
G21 G90 G17 G94
G53 G0 Z-10
M0
S18000 M3
G4 P3
G0 X10 Y10
G0 Z10
…
M5
G53 G0 Z-10
G0 X0 Y0
M30
```

The post-processor, rapid speed (for the time estimate), safe Z, clearance Z, comments and G2/G3 arcs are set in the Project tab, Machine block. The file gets the extension the post-processor expects ( `.nc` , `.tap` , `.ngc` ).

**Curves and G2/G3 arcs.** COPO does not send Bézier curves to the machine as they are: toolpaths are computed as short segments, then, when “G2/G3 arcs” is ticked, circular portions are recognised and written as true G2/G3 arcs (shorter file, smoother motion). A circle drawn with the Circle tool or a vector smoothed in *Arcs* mode gives exact arcs; a free-form curve is approximated by a series of arcs and lines, within tolerance.

> Always check the G-code with a dry run, tool above the stock, before cutting.

## Units: millimetres or inches

In the Project tab, Units block, choose the **Length units** (millimetres or inches) and the **Feed units** (mm/min, mm/s or in/min). The choice applies everywhere: drawing, rulers, grid (1/16, 1/8, 1/4 and 1 inch), coordinates, dimensions, tools, toolpaths and setup sheet. In inches, one option shows values as fractions (to 1/64) and another writes the G-code in inches ( `G20` ) for controllers set up that way.

The project is always saved in millimetres: you can switch units at any time without distorting anything, and a project opens correctly for someone working in the other unit. The choice is remembered in the browser.

## Setup sheet

The **Setup sheet…** button in the G-code tab (also in the Machining menu) prepares a two-page setup sheet for the workshop. Enter the part name, the

material and your notes; the sheet contains:

- the stock, X0 Y0 origin, Z zero, flip and post-processor;
- a summary: estimated time, maximum depth, machined extent from the origin, cutting length and removed material volume;
- the 2D view of each side with toolpaths, origin and stock dimensions, and two dimetric views of the simulation (side 1 and side 2);
- the tool loading order, then the toolpath table: operation, tool, depth, passes, allowance, entry, tabs, feed, plunge and time;
- the points to watch it detects (cut through the stock, cut-out without tabs, pass too deep, job outside the stock, hidden toolpaths);
- a checklist before starting, a notes area and sign-off boxes.

**Save as PDF** (in the sheet window, or straight from **File › Save the setup sheet as PDF** without opening it) creates an A4 PDF file, split into pages without cutting tables or pictures, ready to print or send. The PDF is a picture of the sheet (text not selectable); for a PDF with text, use **Print** then the browser's "Save as PDF". You can also keep the HTML file with the G-code. The sheet settings are saved with the project.

# Install and use offline

- **Online**: open the GitHub Pages site of the COPO repository. Nothing to install.
- **Offline**: download the complete COPO folder ("Code › Download ZIP" on GitHub) and unzip it. Open `index.html` with a double-click: everything is already in the `lib` folder (calculation libraries and interface fonts) and in the `fonts` folder. Nothing else to install or download.
- **Interface fonts**: the `lib/fonts` folder holds the COPO fonts (Source Sans 3, JetBrains Mono, Barlow Condensed). Offline, the interface, the setup sheet and this guide keep the same look.
- **Engraving fonts**: put your `.ttf`, `.otf` or `.woff` files in a `fonts` folder next to COPO.html and open it once from the File menu.

COPO runs on Windows, macOS and Linux, in a recent Chrome, Edge or Firefox. The list of installed fonts is only available in Chrome and Edge.

# Keyboard shortcuts

| KEY | ACTION | KEY | ACTION |
|---|---|---|---|
| `V` | Select | `Ctrl` + `Z` / `Y` | Undo / Redo |
| `N` | Nodes | `Ctrl` + `S` / `O` | Save / Open a project |
| `H` or `Space` | Pan the view | `Ctrl` + `D` | Duplicate |
| `R` `E` `C` | Rectangle, ellipse, circle | `Ctrl` + `A` | Select all |
| `O` | Polygon and star | `Ctrl` + `G` | Group ( `Shift` : ungroup) |
| `P` `A` `T` | Pen, arc, text | `Esc` / right-click | Back to selection, deselect |
| `K` `F` | Chamfer, fillet | `Del` | Delete the selection or node |
| `X` `U` | Trim, cut | Arrows | Move by 1 mm ( `Shift` : 10 mm) |
| `M` `D` | Measure, dimension | `Esc` | Cancel the current path, deselect |

On a Mac, `Cmd` replaces `Ctrl`.

# Report a bug

**Help › Report a bug…** menu: describe the problem, then click **Open the report on GitHub**. The report arrives pre-filled with technical information (browser, units, stock, toolpaths, recent errors). The workspace screenshot is copied to the clipboard: paste it into the report with `Ctrl` + `V`. If you tick “Attach the project”, drag the downloaded file into the report. A free GitHub account is enough. Ideas for improvements are welcome too at github.com/gbulleryahen-hash/copo/issues.

**Help › Suggest an improvement…** opens the same window for an idea: describe the feature you want and what it would be used for; it reaches GitHub with the *enhancement* label. Both carry the COPO **version number** (also shown in **Help › About**): 1.*n*, where *n* goes up with every published update.

# Licence

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Libraries used, bundled in the `lib` folder: Clipper (Boost Software License), opentype.js (MIT), d3-delaunay and delaunator (ISC).

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